

Requirement Elicitation Tools and Techniques

RequisitePro

The IBM Rational RequisitePro solution is a widely used and familiar Microsoft word tool to ease requirements based on use case model for software development teams focus on improvement of the objectives maximize the quality standard of the object and minimize the threats before the handover of the system.

<https://www.ibm.com/support/pages/rational-requisitepro-713>

<https://www-112.ibm.com/software/howtobuy/softwareandservices/passportadvantage/>

Objectiver

The tool allows stakeholder to take global overview of the system and all the models which demonstrating the system. The tool has possibility to draw the diagram and define concepts of the system and relationship over those concepts. Define models and diagrams are explaining using text document, these documents help in the gathering requirement of the system.

<https://objectiver.com/index.php?id=86>

CaseComplete

The Casecomplete tool is used to edit the textual portion of use cases and provide the ability to create the various types of diagrams including use case diagrams.

<https://casecomplete.com/try>

RMTrack:

The tool is designed with two primary features. First, it allows its users to create their documents as they usually do while adjusting to their needs. Secondly, it keeps the important information in the document themselves. RMTrak allows managing the development projects requirements.

<https://www.rmtrack.com/>

Optimal Trace

Optimal Trace helps requirement engineers to get improved requirements and deliver applications that are precisely aligned with business need.

Optimal trace permits integrating with the outer design tools.

Optimal trace also provides document profiles facility containing the requirements document templates and allows one to create profile and templates.

<https://www.volere.org/tools/optimal-trace/>

DOORS

Dynamic Object Oriented Requirements System (DOORS) is used for traceability management. DOORS tool is used as document management software.

<https://www.ibm.com/docs/en/engineering-lifecycle-management-suite/doors/9.7.0?topic=overview-doors>

Analyst Pro

Analyst Pro is a management tool which facilitate you to deal with the vision of the project. Manage scope of the system using classification of the requirements and artifacts.

<https://analysttool.com/>

Table 1. Assesment of Software Requirement Tools.

Tools/Attributes	Glossary	Templates	Traceability	Tool Integration	Documentation Support	Graphical Representation	Checklist	Scalability
RationalRequisitePro	✗	✓	✓	✓	✓	✓	✗	✗
Objectiver	✓	✓	✓	✓	✓	✓	✓	✗
CaseComplete	✓	✓	✓	✗	✓	✓	✗	✗
RMTrak	✗	✓	✓	✓	✓	✓	✗	✓
Optimal Trace	✗	✓	✓	✓	✓	✓	✗	✗
Analyst Pro	✗	✓	✓	✓	✓	✓	✗	✓
DOORS	✗	✓	✓	✓	✓	✓	✗	✓
GMARC	✗	✓	✓	✗	✓	✓	✗	✗

Card Sorting

Card sorting is a technique in which cards are provided to the client according to the name of domains entity. Card sorting shows how much customer has knowledge of the domain.

Card sorts typically consist of the researcher creating approximately 60 index cards (3"x5"), on which a description of domain entities is printed.

Respondents sort the cards into groups/categories and explain the criteria they use for sorting, and the names they assign to groups.

Card Sorting

1. Card sorts can be used to investigate respondent's recall knowledge of the domain entity
2. Card sorts are a useful technique to distinguish between high and low level problems.
3. Card sorts offer more insights into the target population's views of the topic.
4. Card sort results can provide an input for another technique.
5. card sorting process can be done relatively quickly, at nominal cost, and is flexible and easy for the researcher to handle

Laddering

- In laddering limited set of standard are asked to the customer. The questions are set in hierarchical order.
- This technique can be divided in 3 parts: creation, reviewing and modification.
- Laddering method is a form of structured interview to elicit stakeholder's goals, aims and values .
- It was first introduced by the clinical psychologists in 1960 to understand the people "score values and beliefs.
- Software developers have adapted the laddering techniques for gather the complex user tacit requirements.

Repertory Grid

The aim of the Repertory grid is to identify the resemblance and dissimilarity among the different domain units.

This technique is used when eliciting the detailed data or requirement from the experts.

Techniques of Requirement Elicitation		Direct/Indirect	Qualitative/Quantitative Data	Communication	Understanding the Domain
Classic/Traditional Techniques	Interviews	Direct	Qualitative Data and Quantitative Data	Single-directional with the exception of interviews	Yes
	Surveys	Indirect	Quantitative Data and Qualitative Data	Single-directional with the exception of interviews	Yes
	Questionnaires	Indirect	Quantitative Data	Single-directional with the exception of interviews	Yes
Cognitive /Analytical Techniques	Card sorting	Indirect	Quantitative Data	Single- and Twodirectional	Yes
	Laddering	Indirect	Qualitative Data & Quantitative Data	Single- and Twodirectional	Yes
	Repertory Grids	Indirect	Qualitative Data& Quantitative Data	Single- and Twodirectional	Yes
Modern and Group Elicitation Techniques	Brain Storming	Direct	Qualitative Data	Twodirectional	Yes
	JAD	Direct	Qualitative Data	Twodirectional	No
	Prototyping	Direct	Qualitative Data	Twodirectional	No
Social Analysis	Ethnography	Direct	Qualitative Data	Single- and Twodirectional	Yes
	Direct Observation	Direct	Qualitative Data	Single- and Twodirectional	Yes
	Passive Observation	Indirect	Qualitative Data	Single- and Twodirectional	Yes

Presentation December 3,2024

Prepare a presentation on any selected Requirement Elicitation tool.

The presentation should be comprehensive ,have all steps of how to use tool with Link to youtube Video.

Presentation should discuss pros and cons of the selected tool.

Report the guideline about cognitive and group elicitation technique.

Design that technique for your repository .